GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Fatty Acid Catabolic Process (GO:0009062	−0.21385593	58	1.816e−08	9.784e−05	ECI2:8 EHHADH:236 ACAA2:300 ACAD11:349 SLC27A4:350 ACOT8:365
Muscle Contraction (GO:0006936)	0.16989190	88	3.770e−08	9.804e−05	TACR2:5 MYH13:31 MYH2:142 MYH8:191 KCNJ12:230 MYH1:421
tRNA Methylation (GO:0030488)	−0.26190903	36	5.459e−08	9.804e−05	TRMT10A:151 TRMT10B:164 FTSJ1:311 THADA:358 TRMT1L:595 TRMO:696
Chemical Synaptic Transmission (GO:00072	0.09806908	255	7.894e−08	1.063e−04	RIMBP2:27 NPTX1:69 SLC18A2:89 GABRR1:104 RPS6KA2:122 NTSR1:133
Regulation Of Heart Rate By Cardiac Cond	0.24580968	39	1.101e−07	1.186e−04	DSG2:212 ISL1:366 CAV1:565 KCNQ1:585 BIN1:694 KCNE1:790
Cardiac Conduction (GO:0061337)	0.21969576	44	4.675e−07	4.198e−04	DSG2:212 ISL1:366 CAV1:565 KCNQ1:585 BIN1:694 KCNE1:790
Fatty Acid Beta−Oxidation (GO:0006635)	−0.20978456	47	6.606e−07	4.449e−04	ECI2:8 ACOX2:88 EHHADH:236 GCDH:273 ACAA2:300 ACAD11:349
tRNA Modification (GO:0006400)	−0.17803272	66	5.825e−07	4.449e−04	TRMT10A:151 FTSJ1:311 PUS10:312 THADA:358 DTWD1:439 TRMO:696
Visual Perception (GO:0007601)	0.14552958	95	9.836e−07	5.888e−04	VAX2:38 GNAT2:92 RGS9BP:114 NXNL2:138 CRYAA:145 TRPM1:163
Steroid Metabolic Process (GO:0008202)	−0.17302671	61	3.015e−06	1.624e−03	CYP17A1:28 CYP11A1:60 CYP26C1:99 CYP1A1:124 CYP2R1:146 DHRS4:234
RNA Methylation (GO:0001510)	−0.17508344	58	4.062e−06	1.990e−03	TRMT10A:151 METTL4:206 FTSJ1:311 THADA:358 TFB2M:661 TRMO:696
Striated Muscle Contraction (GO:0006941)	0.18222257	52	5.563e−06	2.498e−03	MYH7B:19 MYH8:191 TNNT3:281 TNNT2:539 KCNQ1:585 GRK2:619
Anterograde Trans−Synaptic Signaling (GO	0.09678607	181	7.535e−06	3.123e−03	NPTX1:69 SLC18A2:89 GABRR1:104 RPS6KA2:122 NTSR1:133 MTNR1B:153
Modulation Of Chemical Synaptic Transmis	0.11783680	118	1.019e−05	3.920e−03	NPTX1:69 TPRG1L:156 CLSTN1:245 GRM6:440 SLC8A3:531 CPLX3:588
Alpha−Amino Acid Catabolic Process (GO:1	−0.23487968	29	1.207e−05	4.334e−03	TAT:122 SARDH:123 KMO:293 ALDH4A1:355 MCCC1:456 HMGCLL1:600
RNA Modification (GO:0009451)	−0.18036546	47	1.909e−05	6.343e−03	FTSJ1:311 DTWD1:439 LCMT2:702 DTWD2:776 PUS3:795 PUS7L:906
Mitochondrial Gene Expression (GO:014005	−0.12325800	100	2.113e−05	6.343e−03	MRPS10:399 LARS2:485 TFB2M:661 MRPL16:692 PTCD3:750 TFAM:796
Sensory Perception Of Light Stimulus (GO	0.12540112	96	2.237e−05	6.343e−03	VAX2:38 GNAT2:92 NXNL2:138 CRYAA:145 TRPM1:163 ARR3:196
tRNA Processing (GO:0008033)	−0.17567427	49	2.124e−05	6.343e−03	FTSJ1:311 DTWD1:439 LCMT2:702 DTWD2:776 PUS3:795 METTL2B:1118
Monocarboxylic Acid Transport (GO:001571	−0.15305869	63	2.696e−05	7.264e−03	SLC5A12:96 SLC16A5:106 SLC27A4:350 ARID4A:369 ATP8B1:449 SLC10A6:589
Mitochondrial Translation (GO:0032543)	−0.12284819	96	3.274e−05	8.400e−03	MRPS10:399 LARS2:485 TSFM:673 MRPL16:692 PTCD3:750 MTIF2:751
Double−Strand Break Repair Via Homologou	−0.11558454	106	4.047e−05	9.911e−03	FANCM:37 SLF1:216 XRCC1:241 MCM9:465 HELQ:537 CDC45:708
Regulation Of Synaptic Transmission, Glu	0.16008349	54	4.779e−05	1.120e−02	TSHZ3:123 TPRG1L:156 CACNG4:402 GRM6:440 DGKI:800 DRD2:847
Non−Motile Cilium Assembly (GO:1905515)	−0.21049228	31	5.021e−05	1.127e−02	ARL13B:170 IFT122:308 C2CD3:371 TOGARAM1:677 CEP89:754 ENKD1:912
Calcium Ion Import Across Plasma Membran	0.20312820	33	5.420e−05	1.168e−02	TRPM1:163 SLC8A3:531 SCN1A:576 SCN2A:716 CACNA1B:965 TRPV3:988
Regulation Of Chromosome Separation (GO:	−0.36542097	10	6.301e−05	1.306e−02	SMC4:156 NCAPD2:342 SMC2:354 NUMA1:712 NCAPH:1664 NCAPG:1795
Cardiac Muscle Cell Action Potential Inv	0.21557543	28	7.913e−05	1.550e−02	SCN1A:576 KCNQ1:585 FGF12:701 KCNE1:790 RYR2:1131 SCN4B:1221
Cellular Response To Oxygen−Containing C	0.06029899	367	8.056e−05	1.550e−02	ADCY1:25 IL37:55 RPTOR:152 SLC25A24:167 TGM2:168 IL36G:204
Adenylate Cyclase−Modulating G Protein−C	0.09277648	145	1.193e−04	2.143e−02	ADGRG7:7 ADGRB1:21 ADCY1:25 GNAT2:92 VIPR1:136 FFAR3:286
Cellular Response To Acetylcholine (GO:1	0.35151708	10	1.186e−04	2.143e−02	CHRM3:250 SLURP2:529 LYPD1:614 CHRND:626 CHRNB2:662 LY6G6D:1574
Neuron Projection Morphogenesis (GO:0048	0.09541954	134	1.414e−04	2.457e−02	ADGRB1:21 DBNL:119 CNTN4:165 DIP2A:242 DVL1:252 IQGAP1:273
Amino Acid Catabolic Process (GO:0009063	−0.17641132	38	1.690e−04	2.738e−02	SARDH:123 KMO:293 ALDH4A1:355 MCCC1:456 AMDHD1:573 HIBCH:657
Fatty Acid Oxidation (GO:0019395)	−0.15903802	47	1.634e−04	2.738e−02	ECI2:8 EHHADH:236 ACAA2:300 ACAD11:349 HACL1:585 ABCD3:841
Sister Chromatid Segregation (GO:0000819	−0.18899065	33	1.728e−04	2.738e−02	CHAMP1:78 SMC4:156 KIF18B:503 ESPL1:562 KIF18A:648 KIFC1:818
Cell Junction Assembly (GO:0034329)	0.11829310	83	1.986e−04	3.057e−02	SHANK2:32 CDH26:59 ITGA6:117 DBNL:119 NRCAM:292 SDK1:383
Adenylate Cyclase−Inhibiting G Protein−C	0.15763554	46	2.184e−04	3.222e−02	CHRM3:250 FFAR3:286 HRH3:318 OPRK1:445 OPRL1:837 DRD2:847
Potassium Ion Transmembrane Transport (G	0.09329398	132	2.213e−04	3.222e−02	KCNV2:20 KCNN3:97 KCNA10:215 KCNJ12:230 KCNQ2:322 KCNS1:372
DNA Repair (GO:0006281)	−0.06455481	274	2.512e−04	3.223e−02	FANCM:37 PARP4:67 MBD4:79 RNF168:87 ERCC6L2:116 RHNO1:138
NADH Dehydrogenase Complex Assembly (GO:	−0.15327488	48	2.412e−04	3.223e−02	ECSIT:98 NDUFS3:376 NDUFAF2:421 NDUFAF1:496 TMEM126A:948 NDUFB8:1145
Mitochondrial Respiratory Chain Complex	−0.15327488	48	2.412e−04	3.223e−02	ECSIT:98 NDUFS3:376 NDUFAF2:421 NDUFAF1:496 TMEM126A:948 NDUFB8:1145

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
HAMAI_APOPTOSIS_VIA_TRAIL_UP	−0.10181764	594	3.391e−17	2.200e−13	SPG11:1 AGGF1:6 PJA2:7 CEP192:41 MBD4:79 NARS1:82
NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO	0.18341469	128	8.135e−13	2.639e−09	NPBWR2:23 CDH26:59 CABLES2:93 DIDO1:102 TCFL5:129 NTSR1:133
MARTENS_TRETINOIN_RESPONSE_UP	0.07997971	676	1.828e−12	3.954e−09	MMEL1:3 COL6A1:11 SHANK2:32 TRIM29:43 PRDM16:63 NPTX1:69
REACTOME_NEURONAL_SYSTEM	0.10230679	383	7.350e−12	1.071e−08	CAMK2B:13 KCNV2:20 ADCY1:25 SHANK2:32 SLC18A2:89 KCNN3:97
REACTOME_MUSCLE_CONTRACTION	0.14513608	187	8.254e−12	1.071e−08	CAMK2B:13 ITPR1:139 MYH8:191 MYBPC2:218 KCNJ12:230 NEB:240
REACTOME_SIGNALING_BY_GPCR	0.07836216	629	2.550e−11	2.757e−08	TACR2:5 TIAM2:12 CAMK2B:13 NPBWR2:23 ADCY1:25 OPN3:62
REACTOME_G_ALPHA_I_SIGNALLING_EVENTS	0.11393493	268	1.496e−10	1.386e−07	CAMK2B:13 NPBWR2:23 ADCY1:25 OPN3:62 GNAT2:92 C3:125
FISCHER_DREAM_TARGETS	−0.06120390	882	9.505e−10	7.709e−07	MTBP:13 FANCM:37 CEP192:41 MDM1:54 DEPDC1:101 WDR76:134
DUTERTRE_ESTRADIOL_RESPONSE_24HR_UP	−0.10073451	308	1.330e−09	9.587e−07	KCNK15:21 TET2:93 WDR76:134 CLSPN:136 ZWILCH:196 STIL:201
NIKOLSKY_BREAST_CANCER_16Q24_AMPLICON	0.25499659	46	2.203e−09	1.429e−06	PIEZO1:26 SPATA2L:208 CPNE7:256 ZC3H18:324 TCF25:388 CHMP1A:682
MIKKELSEN_MEF_HCP_WITH_H3K27ME3	0.07425225	555	2.653e−09	1.565e−06	CAMK2B:13 ADGRB1:21 SLC18A2:89 AATK:113 NTSR1:133 CNTN4:165
REACTOME_DEVELOPMENTAL_BIOLOGY	0.05769274	937	2.981e−09	1.612e−06	COL6A1:11 COL6A2:40 KRT80:64 TGM1:99 RPS6KA2:122 DSG3:130
REACTOME_INFECTIOUS_DISEASE	0.06069607	809	5.728e−09	2.859e−06	ADCY1:25 SH3GL3:50 ACE2:51 NOD1:106 C3:125 ITPR1:139
NIKOLSKY_BREAST_CANCER_7P22_AMPLICON	0.27060706	38	7.860e−09	3.469e−06	LFNG:222 MICALL2:496 INTS1:870 CYP2W1:951 FAM20C:1024 ZFAND2A:1085
MIKKELSEN_NPC_HCP_WITH_H3K27ME3	0.09250206	331	8.019e−09	3.469e−06	BNC1:15 ADGRB1:21 PRDM16:63 VIPR1:136 GATA5:285 TMEM151A:302
RODRIGUES_THYROID_CARCINOMA_POORLY_DIFFE	−0.06819354	596	1.543e−08	6.256e−06	AGGF1:6 LEPROTL1:29 CCPG1:36 SCD:56 APC:107 WDR76:134
NIKOLSKY_BREAST_CANCER_12Q24_AMPLICON	0.41628636	15	2.375e−08	9.062e−06	EP400:4 SFSWAP:6 GOLGA3:291 PUS1:415 P2RX2:487 MMP17:524
DACOSTA_UV_RESPONSE_VIA_ERCC3_COMMON_DN	−0.07589292	449	4.006e−08	1.444e−05	PJA2:7 ZNF292:15 AKAP9:19 MAN2A1:27 DST:35 PARG:38
MEISSNER_NPC_HCP_WITH_H3K4ME2_AND_H3K27M	0.08762926	331	4.650e−08	1.588e−05	BNC1:15 ADGRB1:21 PRDM16:63 VIPR1:136 TMEM151A:302 CADM3:305
DACOSTA_UV_RESPONSE_VIA_ERCC3_DN	−0.05592323	824	6.187e−08	2.007e−05	SPG11:1 TMEM131L:2 PJA2:7 ZNF292:15 AKAP9:19 MAN2A1:27
WP_GPCRS_CLASS_A_RHODOPSINLIKE	0.10667564	211	9.737e−08	2.872e−05	NPBWR2:23 OPN3:62 NTSR1:133 MTNR1B:153 OPN1LW:175 BDKRB1:214
NIKOLSKY_BREAST_CANCER_21Q22_AMPLICON	0.41179702	14	9.557e−08	2.872e−05	COL6A1:11 COL6A2:40 SPATC1L:56 DIP2A:242 ADARB1:330 LSS:432
REACTOME_INNATE_IMMUNE_SYSTEM	0.05352342	865	1.151e−07	3.120e−05	NOD1:106 ANPEP:115 DBNL:119 RPS6KA2:122 C3:125 ITPR1:139
KEGG_TIGHT_JUNCTION	0.13741202	125	1.154e−07	3.120e−05	MYH7B:19 MYH13:31 MYH2:142 AFDN:173 MYH8:191 CLDN22:221
REACTOME_GPCR_LIGAND_BINDING	0.07728889	398	1.340e−07	3.479e−05	TACR2:5 NPBWR2:23 OPN3:62 C3:125 UTS2R:132 NTSR1:133
DAZARD_RESPONSE_TO_UV_NHEK_DN	−0.09082150	282	1.645e−07	4.104e−05	TMEM131L:2 AKAP9:19 DST:35 AFF1:80 TUT4:104 PPRC1:131
BLANCO_MELO_BRONCHIAL_EPITHELIAL_CELLS_I	−0.11873872	163	1.751e−07	4.207e−05	WDR76:134 CLSPN:136 RHNO1:138 ORC1:172 IRAG2:192 EPN3:227
WP_STRIATED_MUSCLE_CONTRACTION_PATHWAY	0.25771699	34	1.995e−07	4.622e−05	MYH8:191 MYBPC2:218 NEB:240 TNNT3:281 TNNT2:539 DES:569
DODD_NASOPHARYNGEAL_CARCINOMA_DN	−0.04432260	1234	2.246e−07	4.857e−05	FANCM:37 GSTO1:39 CEP192:41 EMILIN2:72 DEPDC1:101 NSD3:108
MARSON_BOUND_BY_E2F4_UNSTIMULATED	−0.05972583	655	2.176e−07	4.857e−05	MTBP:13 FANCM:37 KIAA0825:40 MDM1:54 TBCCD1:58 UBAP2:65
BENPORATH_ES_WITH_H3K27ME3	0.04857501	982	3.294e−07	6.891e−05	PLEC:1 CAMK2B:13 BNC1:15 NKPD1:28 VAX2:38 IGFBP1:46
REACTOME_CLASS_A_1_RHODOPSIN_LIKE_RECEPT	0.08627174	296	3.525e−07	6.891e−05	TACR2:5 NPBWR2:23 OPN3:62 C3:125 UTS2R:132 NTSR1:133
REACTOME_CARDIAC_CONDUCTION	0.13446325	120	3.717e−07	6.891e−05	CAMK2B:13 ITPR1:139 KCNJ12:230 ATP1A4:389 CACNG4:402 ATP2A3:510
NUYTTEN_EZH2_TARGETS_DN	−0.04942909	938	3.668e−07	6.891e−05	PJA2:7 MTBP:13 LEPROTL1:29 WNK4:47 ABHD11:68 DEPDC1:101
JOHNSTONE_PARVB_TARGETS_3_DN	−0.05385792	784	3.541e−07	6.891e−05	MTBP:13 MAN2A1:27 CCPG1:36 MDM1:54 UBAP2:65 PHF20:91
KEGG_NEUROACTIVE_LIGAND_RECEPTOR_INTERAC	0.09139775	261	3.931e−07	7.084e−05	TACR2:5 NPBWR2:23 GABRR1:104 UTS2R:132 NTSR1:133 VIPR1:136
REACTOME_NEUROTRANSMITTER_RECEPTORS_AND_	0.10762113	183	5.338e−07	9.361e−05	CAMK2B:13 ADCY1:25 GABRR1:104 RPS6KA2:122 KCNJ12:230 GABRR3:311
MIKKELSEN_MCV6_HCP_WITH_H3K27ME3	0.07117631	420	6.196e−07	1.033e−04	CAMK2B:13 FAM83G:16 ADGRB1:21 SLC18A2:89 L1CAM:131 NTSR1:133
RODRIGUES_THYROID_CARCINOMA_ANAPLASTIC_U	−0.05833447	634	6.211e−07	1.033e−04	WDR36:69 DEPDC1:101 APC:107 SMC4:156 ZWILCH:196 STIL:201
REACTOME_KERATINIZATION	0.15578602	85	6.984e−07	1.133e−04	KRT80:64 TGM1:99 DSG3:130 KRT13:157 TGM5:166 KRT19:206

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
HIV Infections	0.05654629	634	1.516e−06	7.438e−03	PLEC:1 IL16:57 ISG20:58 LPIN1:84 VIPR1:136 IL18:141
Tetany	−0.30936826	21	9.254e−07	7.438e−03	CLDN16:25 TRPM6:45 JMJD1C:190 SLC12A3:472 COMT:646 SLC12A1:705
Glycinuria	−0.35951644	13	7.192e−06	2.352e−02	SLC6A20:209 ALDH4A1:355 PCCA:842 PRODH:893 ACADM:1372 SLC36A2:1597
CATARACT, AUTOSOMAL DOMINANT	0.39546196	10	1.490e−05	2.436e−02	CRYAA2:76 CRYAA:145 GJA8:228 CRYBB2:380 BFSP2:560 MIP:656
Congenital cerebral hernia	−0.18581002	46	1.315e−05	2.436e−02	CRB1:31 WDPCP:323 RDH12:378 TMEM67:515 ARHGAP31:568 CEP290:606
Mouth Neoplasms	0.12317614	107	1.107e−05	2.436e−02	UPP1:45 IL18:141 PTK2:144 HMGCS2:158 CDKN2B:335 CAV1:565
HYPERGLYCINURIA (disorder)	−0.31797264	15	2.013e−05	2.542e−02	SLC6A20:209 ALDH4A1:355 PCCA:842 PRODH:893 ACADM:1372 SLC36A2:1597
Nuclear cataract	0.21751187	32	2.073e−05	2.542e−02	CRYAA:145 TGM2:168 HSF4:179 GJA8:228 CRYBB2:380 MIP:656
22q11 partial monosomy syndrome	−0.36750784	11	2.439e−05	2.658e−02	JMJD1C:190 COMT:646 HIRA:1086 TBX1:1435 PI4KA:1707 GP1BB:2115
Abnormality of the tonsils	−0.35741233	11	4.053e−05	3.058e−02	JMJD1C:190 COMT:646 HIRA:1086 TBX1:1435 GP1BB:2115 IDUA:2808
Epidermolysis Bullosa Simplex	0.26066410	21	3.559e−05	3.058e−02	PLEC:1 ITGB4:9 KRT80:64 TGM5:166 DES:569 KRT14:597
Hypocalcemia	−0.17798902	45	3.648e−05	3.058e−02	CLDN16:25 TRPM6:45 SLC4A1:139 JMJD1C:190 IFT122:308 COMT:646
Primary microcephaly	−0.11618317	105	4.007e−05	3.058e−02	FANCM:37 TRMT10A:151 WDFY3:154 ORC1:172 STIL:201 ERCC6:237
Night blindness, congenital stationary	0.22738136	25	8.341e−05	5.455e−02	TRPM1:163 GPR34:244 PDE6B:339 GRM6:440 FGFR2:542 NYX:549
Complete congenital stationary night bli	0.30411970	14	8.164e−05	5.455e−02	TRPM1:163 PDE6B:339 GRM6:440 FGFR2:542 NYX:549 GPR179:683
Anemia, hereditary spherocytic hemolytic	−0.49058568	5	1.452e−04	6.049e−02	ANK1:32 EPB42:77 SLC4A1:139 SPTA1:140 SPTB:372 NA
Cataract, Central Saccular, With Sutural	0.44610166	6	1.542e−04	6.049e−02	GJA8:228 CRYBB2:380 BFSP2:560 MIP:656 CRYBA1:858 CRYGS:2341
Asthma	0.03580659	1060	1.095e−04	6.049e−02	MMEL1:3 TACR2:5 SFSWAP:6 ITGB4:9 IL37:55 IL16:57
Dysseborrheic dermatitis	−0.27613601	16	1.314e−04	6.049e−02	JMJD1C:190 COMT:646 HIRA:1086 TBX1:1435 BTD:1826 GP1BB:2115
Familial intrahepatic cholestasis of pre	−0.44760351	6	1.465e−04	6.049e−02	NR1H4:417 ATP8B1:449 ABCB11:584 VCAM1:613 ABCB4:1135 TNF:1749
Impaired T cell function	−0.27109497	17	1.091e−04	6.049e−02	JMJD1C:190 COMT:646 UMPS:1069 HIRA:1086 TBX1:1435 CR2:1842
Mildly elevated creatine phosphokinase	0.23078718	23	1.279e−04	6.049e−02	COL6A1:11 COL6A2:40 NEB:240 MYH14:331 GNE:732 LMNA:832
Occipital myelomeningocele	−0.36763360	9	1.340e−04	6.049e−02	JMJD1C:190 COMT:646 HIRA:1086 TBX1:1435 GP1BB:2115 UFD1:3236
Oguchi disease	0.27339559	16	1.532e−04	6.049e−02	TRPM1:163 PDE6B:339 GRM6:440 FGFR2:542 NYX:549 GPR179:683
Spherocytosis	−0.49058568	5	1.452e−04	6.049e−02	ANK1:32 EPB42:77 SLC4A1:139 SPTA1:140 SPTB:372 NA
Neuromuscular Diseases	0.11407243	91	1.726e−04	6.512e−02	PLEC:1 MYH2:142 NEB:240 DES:569 CHRND:626 TIMP2:709
Waddling gait	0.14862499	53	1.838e−04	6.678e−02	L1CAM:131 NEB:240 BICD2:497 CANT1:617 BIN1:694 TGFB1:745
Familial benign neonatal epilepsy	0.31005673	12	2.002e−04	6.771e−02	KCNQ2:322 CHRNA4:373 KCNQ3:522 SCN2A:716 DNAH8:882 SCN8A:1046
Infection	0.05484813	396	1.975e−04	6.771e−02	PLEC:1 ACE2:51 IL16:57 NOD1:106 ANPEP:115 ITGA6:117
NIGHT BLINDNESS, CONGENITAL STATIONARY,	0.30546488	12	2.486e−04	7.397e−02	TRPM1:163 PDE6B:339 GRM6:440 NYX:549 GPR179:683 SAG:1079
Hypochloremia (disorder)	−0.43043580	6	2.608e−04	7.397e−02	CLCNKB:5 CLCNKA:9 SLC12A1:705 KCNJ1:924 SLC26A3:1141 BSND:3912
Nuclear non−senile cataract	0.19656783	29	2.496e−04	7.397e−02	CRYAA:145 GJA8:228 CRYBB2:380 MIP:656 LMNA:832 FTO:835
Posterior embryotoxon	−0.19138161	31	2.272e−04	7.397e−02	JMJD1C:190 PEX1:431 COMT:646 PEX6:716 PEX16:1003 HIRA:1086
Seborrheic dermatitis	−0.23002170	21	2.639e−04	7.397e−02	JMJD1C:190 NFE2L2:232 COMT:646 HIRA:1086 TBX1:1435 TNF:1749
Serum chloride level decreased (finding)	−0.43043580	6	2.608e−04	7.397e−02	CLCNKB:5 CLCNKA:9 SLC12A1:705 KCNJ1:924 SLC26A3:1141 BSND:3912
Congenital cataract	0.11097680	89	3.014e−04	8.214e−02	CRYAA2:76 CRYAA:145 HSF4:179 GJA8:228 CRYGA:297 ATF6:358
Color Blindness, Red	0.46162809	5	3.503e−04	8.217e−02	GNAT2:92 OPN1LW:175 ATF6:358 TEX28:467 PDE6H:1882 NA
Primary hyperoxaluria, type I	−0.28637195	13	3.504e−04	8.217e−02	DAO:373 AGXT:755 HAO2:885 GRHPR:1397 PTHLH:1586 OR10A4:3507
Skin Diseases, Genetic	0.17219281	36	3.518e−04	8.217e−02	ITGB4:9 IGFBP3:86 TGM1:99 ITGA6:117 KRT14:597 ECM1:723
Asymmetric crying face association	−0.28867379	13	3.139e−04	8.217e−02	JMJD1C:190 COMT:646 HIRA:1086 TBX1:1435 GP1BB:2115 CASP3:2896

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0003635 abnormal synaptic transmissio	0.07338461	424	5.093e−07	2.424e−04	CAMK2B:13 ADCY1:25 AFF2:70 SLC18A2:89 L1CAM:131 NTSR1:133
MP0000598 abnormal liver morphology	−0.05935396	471	2.033e−05	4.838e−03	MAN2A1:27 ANK1:32 ANGPTL3:76 INVS:118 CYP1A1:124 PNLIP:186
MP0000609 abnormal liver physiology	−0.05683615	374	2.406e−04	1.628e−02	MAN2A1:27 FFAR1:64 ANGPTL3:76 INVS:118 CYP1A1:124 PLIN2:160
MP0005365 abnormal bile salt	−0.21173315	26	1.911e−04	1.628e−02	PNLIP:186 ALB:228 EHHADH:236 ATP8B1:449 ABCB11:584 CYP7A1:667
MP0001970 abnormal pain threshold	0.08672776	151	2.736e−04	1.628e−02	ADCY1:25 AFF2:70 GABRR1:104 TSHZ3:123 NTSR1:133 BDKRB1:214
MP0009643 abnormal urine homeostasis	−0.07173717	227	2.487e−04	1.628e−02	SLC14A2:4 CLCNKA:9 CLDN16:25 MAN2A1:27 SLC4A1:139 LEPR:195
MP0000678 abnormal parathyroid gland	−0.23149065	21	2.452e−04	1.628e−02	CHRD:229 GCM2:798 TBX1:1435 CASR:1461 CYP27B1:1477 PTH:1558
MP0003252 abnormal bile duct	−0.23838516	22	1.111e−04	1.628e−02	PKHD1:313 NR1H4:417 ATP8B1:449 ABCB11:584 P2RY13:979 ABCB4:1135
MP0003868 abnormal feces composition	−0.20497470	25	3.976e−04	2.103e−02	APC:107 PNLIP:186 ALB:228 CYP7A1:667 SLC10A2:911 ABCB4:1135
MP0002135 abnormal kidney morphology	−0.04905499	450	5.609e−04	2.670e−02	MAN2A1:27 ANK1:32 TET2:93 INVS:118 SLC4A1:139 KCP:149
MP0002136 abnormal kidney physiology	−0.06061764	279	6.387e−04	2.764e−02	SLC14A2:4 CLCNKA:9 CLDN16:25 MAN2A1:27 WNK4:47 SLC4A1:139
MP0005636 abnormal mineral homeostasis	−0.08345471	134	9.528e−04	3.272e−02	CLDN16:25 ANK1:32 TRPM6:45 WNK4:47 CYP11A1:60 SLC4A1:139
MP0002063 abnormal learning/memory/cond	0.05246758	356	9.225e−04	3.272e−02	ADCY1:25 AFF2:70 SLC18A2:89 L1CAM:131 ARC:134 RAI1:211
MP0002272 abnormal nervous system	0.06952962	195	9.622e−04	3.272e−02	PLEC:1 LPIN1:84 GABRR1:104 ARC:134 RAI1:211 CHRM3:250
MP0005253 abnormal eye physiology	0.09935263	91	1.134e−03	3.600e−02	RGS9BP:114 TRPM1:163 GJA8:228 RCVRN:364 ISL1:366 BFSP2:560
MP0005551 abnormal eye electrophysiolog	0.07897478	139	1.457e−03	4.336e−02	NPTX1:69 GNAT2:92 GABRR1:104 C3:125 TRPM1:163 ARR3:196
MP0009745 abnormal behavioral response	0.06520197	204	1.558e−03	4.361e−02	UPP1:45 SLC18A2:89 MBP:316 KCNQ2:322 GAL:352 SLC6A2:362
MP0005332 abnormal amino acid	−0.08373938	116	2.002e−03	5.293e−02	SLC6A6:119 LEPR:195 ARG1:329 GNMT:356 DAO:373 UPK3A:434
MP0002085 abnormal embryonic tissue	−0.03642038	650	2.570e−03	6.439e−02	TRPM6:45 NSD1:61 ADAMTS20:97 ECSIT:98 APC:107 INTU:127
MP0005085 abnormal gallbladder physiolo	−0.17203749	25	2.953e−03	6.818e−02	NR1H4:417 ATP8B1:449 ABCB11:584 CYP7A1:667 SLC10A2:911 P2RY13:979
MP0002148 abnormal hypersensitivity rea	0.08563049	102	3.008e−03	6.818e−02	TNFRSF4:164 TBXA2R:325 SSTR4:393 IL21R:532 CD40:574 PTGER4:748
MP0003195 calcinosis	−0.28482343	8	5.298e−03	1.146e−01	SLC12A1:705 AGXT:755 CASR:1461 ABCC6:1481 ENPP1:2441 SPP1:4486
MP0002064 seizures	0.04989006	253	7.296e−03	1.510e−01	DBNL:119 ITPR1:139 RAI1:211 MBP:316 KCNQ2:322 ADARB1:330
MP0003453 abnormal keratinocyte physiol	0.10007935	58	8.647e−03	1.646e−01	ITGB4:9 PTK2:144 IKBKG:455 PKP3:466 MAP3K6:486 DSC3:519
MP0004019 abnormal vitamin homeostasis	−0.11946817	41	8.302e−03	1.646e−01	CLDN16:25 KCP:149 PLIN2:160 SLC23A1:202 TGIF1:477 CYP7A1:667
MP0002572 abnormal emotion/affect behav	0.04332041	290	1.295e−02	2.370e−01	ADCY1:25 SLC18A2:89 L1CAM:131 ARC:134 ABCA2:209 RAI1:211
MP0010094 abnormal chromosome stability	−0.09163034	61	1.371e−02	2.417e−01	USP1:226 ATR:261 AICDA:594 CENPE:684 PARP2:688 BUB1B:952
MP0002139 abnormal hepatobiliary system	−0.24866872	8	1.492e−02	2.536e−01	ALDH1A1:389 FABP1:985 ABCB4:1135 NR1H3:2031 ABCG8:2072 INSIG1:9275
MP0000313 abnormal cell death	−0.03320195	456	1.882e−02	3.089e−01	PARG:38 MBD4:79 LAMA1:100 APC:107 DAXX:155 STIL:201
MP0000343 altered response to	0.09390035	52	1.960e−02	3.109e−01	TGM2:168 NT5E:379 POSTN:788 MMP2:805 MAP3K5:921 SOD1:1057
MP0001919 abnormal reproductive system	−0.10477914	40	2.221e−02	3.410e−01	TRPM6:45 PTGFR:246 WT1:720 RAB27A:1138 CAMK4:1552 LGR4:1965
MP0001177 atelectasis	0.08839748	54	2.518e−02	3.745e−01	TSHZ3:123 DOCK1:151 SYNE1:295 SFTPB:301 KCNQ2:322 PDGFA:351
MP0001293 anophthalmia	−0.09338195	47	2.726e−02	3.932e−01	HESX1:129 CHRD:229 AP3B1:703 CNBP:1333 BMP4:1548 RAB23:2060
MP0002084 abnormal developmental patter	−0.03407313	359	3.077e−02	4.069e−01	NSD1:61 ECSIT:98 LAMA1:100 APC:107 INVS:118 INTU:127
MP0005319 abnormal enzyme/ coenzyme	−0.04840441	172	3.052e−02	4.069e−01	CYP1A1:124 NFE2L2:232 GNPTAB:351 SMN1:353 GNMT:356 NR1H4:417
MP0003186 abnormal redox activity	−0.08864407	50	3.070e−02	4.069e−01	NFE2L2:232 SMN1:353 WRN:768 XPC:866 RALBP1:974 CBS:1315
MP0003656 abnormal erythrocyte physiolo	−0.13124596	22	3.334e−02	4.290e−01	MAN2A1:27 ANK1:32 SLC4A1:139 NFE2L2:232 AQP9:564 NEK1:644
MP0008058 abnormal DNA repair	−0.09818619	39	3.433e−02	4.301e−01	APC:107 ERCC6:237 PARP2:688 TEX15:933 POLK:1450 POLL:1630
MP0010234 abnormal vibrissa follicle	0.22987110	7	3.527e−02	4.305e−01	PKP3:466 SPINK5:1793 PTEN:2636 CTNNB1:3546 CHUK:6858 ST14:6858
MP0002734 abnormal mechanical nocicepti	0.07260483	69	3.794e−02	4.515e−01	ADCY1:25 GABRR1:104 TSHZ3:123 L1CAM:131 BDKRB1:214 STOML3:439

tissue_specific Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
EBV.lymphocyte	−0.056978311	672	1.118e−06	3.020e−05	TMEM131L:2 MTBP:13 IL23R:75 DEPDC1:101 WDR76:134 CLSPN:136
liver	−0.080585910	328	7.896e−07	3.020e−05	MOGAT1:23 PBLD:71 ANGPTL3:76 ACOX2:88 ACOT12:111 TAT:122
cerebral.cortex	0.130613224	107	3.405e−06	6.129e−05	ADGRB1:21 ADGRA1:312 RCVRN:364 KCNS1:372 C1QL2:385 FAM163B:431
pancreas	−0.134935470	94	6.743e−06	9.102e−05	CPB1:33 TMED6:62 SLC43A1:148 KCNK16:185 PNLIP:186 EGF:251
leg.skin	0.084798152	217	2.038e−05	2.202e−04	BNC1:15 FAM83G:16 VSIG8:22 NKPD1:28 TRIM29:43 IL37:55
suprapubic.skin	0.084003699	194	6.404e−05	5.764e−04	BNC1:15 NKPD1:28 IL37:55 KRT80:64 DSG3:130 CLEC2A:148
Brodmann.area.9	0.082710549	182	1.359e−04	1.048e−03	FRMPD4:100 TMEM132D:290 NRCAM:292 KCNS1:372 NETO1:425 HS3ST2:508
esophagus.mucosa	0.070777719	220	3.437e−04	2.320e−03	BNC1:15 LEXM:71 TGM1:99 FAM83H:110 DSG3:130 CAPN14:140
skeletal.muscle	0.067367291	239	3.902e−04	2.341e−03	IGFN1:67 SMTNL2:111 RGS9BP:114 MYH2:142 MYH8:191 FSD2:217
vagina	0.095216530	91	1.771e−03	9.564e−03	BNC1:15 LEXM:71 SLC18A2:89 DSG3:130 KRT13:157 SLURP1:246
thyroid	−0.073321937	151	2.017e−03	9.901e−03	CLCNKA:9 KCNJ16:50 PRTG:137 WDR72:168 PLA2R1:182 PGF:198
cerebellum	0.040829029	376	7.604e−03	3.422e−02	RNF112:17 GLRA4:47 L1CAM:131 NOS1AP:154 ZBTB46:160 CRTAM:188
cortex.kidney	−0.059831653	114	2.816e−02	1.170e−01	CLCNKB:5 CLDN16:25 SLC5A12:96 TINAG:303 PKHD1:313 OGDHL:367
hypothalamus	0.067806993	66	5.753e−02	2.219e−01	SMIM17:535 PIRT:991 SYT6:1019 GABRG3:1266 DIRAS3:1547 MC4R:1549
left.ventricle	0.062826474	68	7.410e−02	2.668e−01	MYH7B:19 FSD2:217 LMOD2:499 TNNT2:539 MYOM2:607 CKMT2:627
sigmoid.colon	0.071833568	50	7.956e−02	2.685e−01	TACR2:5 NTSR1:133 ROR2:338 DES:569 GREM2:679 CNN1:932
adrenal.gland	−0.046712384	108	9.516e−02	2.705e−01	SLC14A2:4 ADGRV1:10 CYP17A1:28 CYP11A1:60 MGST1:491 MC2R:540
ectocervix	−0.097638791	25	9.140e−02	2.705e−01	RIPOR3:587 PAPPA2:689 PTHLH:1586 LRFN5:2218 KRT34:2488 FGF7:2574
prostate	0.079936712	38	8.870e−02	2.705e−01	CDH26:59 P2RX2:487 ACP3:568 ANO7:655 NR2E3:856 HOXB13:1386
fallopian.tube	−0.065888718	52	1.011e−01	2.728e−01	GREB1L:153 ADAMTS3:199 GMNC:304 GAS2L2:580 CCDC17:724 LDLRAD1:1260
substantia.nigra	−0.104260195	12	2.113e−01	5.434e−01	DBX2:693 CHRM5:847 RPE65:3049 C10orf105:4084 SPX:5678 KCNE5:5877
amygdala	0.097204693	9	3.127e−01	5.448e−01	NEUROD6:775 SLC17A7:2162 PTPRZ1:3271 NEUROD2:6858 PCDH15:6858 SLC38A8:6858
aorta	−0.024522143	154	2.971e−01	5.448e−01	KCNK15:21 HMCN1:53 VCAN:508 FRZB:849 NAV2:1130 ITGB1:1167
bladder	−0.060760961	24	3.033e−01	5.448e−01	UPK3A:434 SNX31:637 UPK1A:3301 CLEC3A:3357 DHRS2:3767 UPK1B:4027
cerebellar.hemisphere	0.015861233	509	2.323e−01	5.448e−01	CAMK2B:13 MINAR1:14 RNF112:17 ADCY1:25 VAX2:38 NPTX1:69
hippocampus.proper	0.100743338	10	2.701e−01	5.448e−01	NEUROD6:775 OLIG2:1671 SLC17A7:2162 NEUROD2:6858 HRK:6858 CABP7:6858
nucleus.accumbens	0.030252102	99	3.005e−01	5.448e−01	JAKMIP1:101 INSYN2A:150 PRKCG:378 CACNG4:402 OPRK1:445 LYPD1:614
spleen	−0.017517095	299	3.044e−01	5.448e−01	PATL2:18 STAB2:181 SCML4:299 GP9:347 SCARF1:407 AKNA:424
subcutaneous.adipose	−0.037638339	72	2.710e−01	5.448e−01	CYP1A1:124 PLIN2:160 PPARG:529 LVRN:874 PFKFB3:1370 LGALS12:1399
testis	0.009674453	1382	2.633e−01	5.448e−01	MMEL1:3 BNC1:15 EPS8L3:18 KCNV2:20 DNAH2:35 GSG1:36
uterus	−0.037586350	61	3.113e−01	5.448e−01	CHRDL2:390 MEIS3:430 ATP2B4:681 WT1:720 TEX15:933 CLEC4F:1531
tibial.artery	0.025789459	109	3.547e−01	5.986e−01	GPR20:237 C1QTNF1:298 ADARB1:330 SDK1:383 CDH6:395 SMAD7:649
breast	0.066699415	15	3.714e−01	5.990e−01	THRSP:553 PLIN1:1373 KRT5:1421 GSC:1731 ABCB5:1818 TFAP2B:3413
coronary.artery	−0.063359064	15	3.958e−01	5.990e−01	SUSD5:1297 SOST:2231 CCN3:2781 MYO18B:3021 MTHFD1L:3164 PCOLCE2:4608
endocervix	0.028821213	72	3.993e−01	5.990e−01	ELFN1:53 ZNF516:68 SLC18A2:89 PWP2:107 NXNL2:138 TGM2:168
lung	−0.021496929	139	3.847e−01	5.990e−01	RTKN2:235 GP9:347 PTPRB:391 ADGRF5:868 HTR3C:1063 SFTA2:1156
blood	−0.013844021	276	4.347e−01	6.345e−01	LY6G6F:73 TREML2:74 EPB42:77 HPSE:112 SLC6A6:119 GCA:125
esophagus.muscularis.mucosa	0.041499968	26	4.644e−01	6.599e−01	SYNM:197 CHRM3:250 TPM1:1337 FAM83D:1820 KCNMB1:1949 NKX6−1:6858
tibial.nerve	−0.014837153	192	4.824e−01	6.679e−01	SECISBP2L:30 MLIP:133 EHBP1:152 LEPR:195 CNTF:225 PTGFR:246
peyers.patch	−0.015383116	167	4.962e−01	6.699e−01	SLC5A12:96 SLC23A1:202 SLC6A20:209 TINAG:303 MYRFL:334 CREB3L3:500